

Style Nursing++: Zero-Shot Compound Weather Transfer with Adaptive Style Nursing and Learned Structural Kernel

Chi Zhang, *Member, IEEE*, Yuchen Li, *Member, IEEE*, Le Wang, *Member, IEEE*, Meng Yuan, *Member, IEEE*, Yuanqi Su, *Member, IEEE*, Li Li, *Member, IEEE*

Abstract—Zero-shot compound weather transfer must express multiple prompt concepts while preserving the geometry and semantics of the source scene. Existing diffusion editors commonly suffer from two failure modes: Style Neglection, where one or more requested concepts remain under-expressed, and Style Over-expression, where stronger style guidance over-edits the scene and damages the original structure. We present *Style Nursing++*, which couples adaptive concept reinforcement with selective structural guidance. *Adaptive Style Nursing* identifies style-bearing concepts, routes each concept to a compatible spatial support, and scales latent updates according to a regional cross-attention deficit. *Layer-Selective Spatial Guidance* anchors source geometry at selected U-Net stages, while *Entropy-Selective Semantic Guidance* preserves informative object relations through concentrated self-attention heads. However, semantic guidance requires additional computation on semantic maps. Inspired by KDE-based attention estimation, we further introduce a *Learned Structural Kernel* (LSK), a Gaussian attention operator trained to reproduce semantic-guided structural responses. At inference, LSK aggregates the current value features and removes the auxiliary semantic-reference trajectory. On BDD100K, Cityscapes, and STREET-6K, our approach outperforms state-of-the-art zero-shot image translation methods across comprehensive quantitative metrics, achieving better structural preservation and style adaptation. Beyond generation quality, our method reduces the mean latency by 28.8 %, computational cost by 20.9 %. These results reveal that the attention patterns required by diffusion guidance can be effectively learned through KDE-inspired kernel operators, reducing the computational overhead of repeated attention-map computation.

Index Terms—Compound weather transfer, diffusion-based image editing, prompt faithfulness, structural preservation, adaptive style control, learned structural kernel.

I. INTRODUCTION

Text-guided diffusion models provide a flexible interface for image editing, yet compound weather transfer remains difficult in the zero-shot setting. A single prompt may request several effects, such as *snow + sunset* or *rain + rainbow*. These effects occupy different spatial supports and emerge at different strengths: snow can cover the entire scene, sunset primarily changes the sky and illumination, and puddles or reflections belong on the road. A successful editor must coordinate these concepts without changing the geometry or semantic organization of the source scene.

This requirement exposes two coupled failure modes. The first is *style under-expression*: an editor preserves the source but weakly renders one or more requested concepts. In Fig. 1(a), for example, Plug-and-Play (PnP) produces snow but largely

misses the requested sunset. Increasing style guidance can recover the missing concept, yet it may cause *structural over-editing*. Fig. 1(b) shows this second failure: PnP introduces fog while changing the foreground vehicle. Compound weather transfer therefore requires enough concept activation to satisfy the prompt and enough structural control to preserve the scene.

We address style under-expression with *Adaptive Style Nursing*. The method identifies style-bearing token spans, assigns each concept to a compatible support—the full image, sky, or road—and measures concept evidence inside that support. A regional deficit compares the current evidence with a concept-specific target. Concepts with larger deficits receive stronger latent updates, while sufficiently expressed concepts receive little or no additional correction. The update is therefore localized and demand-driven.

Structural preservation requires a complementary mechanism because intermediate diffusion representations mix geometry and appearance to different degrees. We introduce *Layer-Selective Spatial Guidance* (LSSG) to align source and edited features at U-Net stages that retain coarse layout and local geometry. We also introduce *Entropy-Selective Semantic Guidance* (ESSG), which selects concentrated self-attention heads from a source-derived semantic reference and aligns their object-level relations. LSSG constrains where scene elements remain, and ESSG constrains how their internal and pairwise relations evolve.

Explicit ESSG requires an auxiliary semantic-reference trajectory at inference. To remove this branch, we train a *Learned Structural Kernel* (LSK) from the structural attention responses produced by the guided trajectory. The kernel predicts current-hidden-state- and timestep-conditioned Gaussian attention weights and applies them to the current value features. This design retains source-conditioned structural aggregation while allowing the edited appearance to evolve through the current values.

Together, these components form *Style Nursing++*, an extension of Style Nursing [1] from uniform token excitation and explicit semantic guidance to region-adaptive concept control, selective structural preservation, and learned structural aggregation. Our contributions are:

- We propose Adaptive Style Nursing, which routes each style concept to a compatible region and adapts its update strength to a regional attention deficit.
- We introduce complementary spatial and semantic constraints. LSSG aligns selected source features across scales,



Fig. 1. Two recurrent failures in text-guided weather transfer. (a) *Style under-expression*: for the *snowy* + *sunset* prompt, Plug-and-Play (PnP) captures snow but weakly represents sunset, whereas *Style Nursing++* activates both concepts while retaining the source layout. The heat maps visualize cross-attention for the *snowy* and *sunset* tokens; warmer colors indicate larger responses. (b) *Structural over-editing*: PnP introduces fog but changes the foreground vehicle, whereas *Style Nursing++* preserves its structure. The lower panels show PCA projections of residual features used by spatial guidance and self-attention relations used by semantic guidance; boxes indicate the matched vehicle region.

and ESSG preserves informative self-attention relations through entropy-based head selection.

- We develop LSK, a current-hidden-state- and timestep-conditioned Gaussian operator that learns the structural aggregation induced by ESSG and replaces the auxiliary semantic-reference trajectory at inference. Experiments show consistent gains over the guidance-only core together with lower latency, compute, and energy.

II. RELATED WORK

A. Diffusion Models and Real-Image Editing

Denoising diffusion probabilistic models formulate image synthesis as the reversal of a gradual noising process [2]. DDIM introduces deterministic non-Markovian sampling and inversion [3], while latent diffusion performs denoising in a compressed autoencoder space and uses cross-attention for text conditioning [4]. Classifier-free guidance (CFG) controls the strength of that conditioning without a separate classifier [5]. Real-image editing additionally requires a source-aligned latent trajectory and constraints that localize the requested change.

Existing methods impose these constraints at different stages. SDEdit starts from a partially noised source [6]; Prompt-to-Prompt reuses or edits cross-attention maps [7]; and Null-text inversion optimizes unconditional embeddings to improve reconstruction [8]. DiffEdit estimates an edit mask from denoising predictions [9], whereas Imagic optimizes a text embedding and fine-tunes the model for each image [10]. Instruction-trained editors, including InstructPix2Pix and InstructDiffusion, learn broad editing behavior from paired or synthesized instructions [11], [12].

A second line of work performs editing without task-specific training or per-image model fine-tuning. Plug-and-Play Diffusion Features injects source features and attention

projections [13]; Pix2Pix-Zero uses cross-attention guidance without paired data [14]; and DiffuseIT separates style and content through diffusion representations [15]. MasaCtrl retrieves source appearance through mutual self-attention [16]. LEDITS++ combines fast inversion with implicit masks and supports simultaneous edits [17]. Free-Prompt-Editing modifies selected self-attention maps using insights from cross- and self-attention analysis [18]. Compound weather transfer adds a spatial allocation problem to these settings: several concepts must be expressed over different supports while traffic-scene geometry remains stable.

B. Prompt Faithfulness and Adverse-Weather Transfer

Text-to-image models may omit a requested entity or bind an attribute to the wrong region. Attend-and-Excite strengthens under-attended subject tokens [19]; CONFORM uses contrastive attention objectives for object separation and attribute binding [20]; and Attention Refocusing imposes cross- and self-attention losses under a specified layout [21]. Semantic-aware CFG assigns different guidance strengths to semantic regions and demonstrates the limits of a single global scale [22]. These methods motivate feedback from intermediate attention, whereas Adaptive Style Nursing focuses on multiple weather concepts and assigns each concept a scene-compatible support.

Classical neural style transfer established the separation of content and appearance. Gatys *et al.* combine feature reconstruction with Gram-matrix statistics [23], and AdaIN matches channel-wise statistics for feed-forward arbitrary style transfer [24]. Adverse-weather synthesis for autonomous driving has often relied on condition-specific translation: density-aware haze synthesis disentangles content and weather style [25], day-to-night translation targets nighttime detector training [26], TPSeNCE improves unpaired rain generation [27],

and RainSD diversifies rain styles in feature space [28]. Recent diffusion methods study when content and style should enter denoising. StyleFM combines frequency manipulation with recursive attention [29], while ACID-Style learns adaptive condition injection through lightweight adapters [30]. Style Nursing introduced zero-shot traffic-scene style transfer and the STREET-6K benchmark using style-token excitation with spatial and semantic guidance [1]. Our work extends that setting to compound prompts, regional deficits, selective structural guidance, and learned structural aggregation.

C. Structure Preservation and Spatial Control

Spatial control can come from an external condition or from the source image itself. ControlNet learns a parallel branch for edges, depth, pose, and segmentation [31]; T2I-Adapter learns lightweight modules for related signals [32]; and FreeControl derives structure and appearance guidance from an arbitrary condition [33]. DiffEditor combines image prompts, regional score guidance, and mixed stochastic–deterministic sampling [34]. StyleAligned shares attention across generated images to transfer a reference style consistently [35]. In contrast, our setting derives control from the source trajectory and the editing prompt without requiring an additional user-provided condition.

Internal representations also encode structure. Local self-similarity descriptors preserve geometric repetition across appearance changes [36], and Vision Transformer features can separate source structure from target appearance [37]. Within diffusion editors, intermediate features carry local layout, whereas self-attention exposes longer-range relations; PnP, MasaCtrl, and Free-Prompt-Editing use different parts of this hierarchy [13], [16], [18]. LSSG and ESSG exploit the same complementarity but select U-Net stages and attention heads rather than copying every source representation.

D. Kernelized Attention and Structural Priors

The Nadaraya–Watson estimator expresses nonparametric regression as normalized kernel-weighted averaging [38], [39]. Related work interprets attention through kernel aggregation. Linear Transformers use feature maps and associativity to obtain linear-time attention [40], Performer approximates the softmax kernel with positive random features [41], and KDE-former connects attention normalization and value aggregation to kernel density estimation [42]. A Gaussian Nadaraya–Watson formulation has also been used to characterize the representational capacity of softmax self-attention [43].

LSK uses this correspondence for structural editing. It learns a current-hidden-state- and timestep-conditioned Gaussian kernel from self-attention responses generated by the semantic-guided trajectory. The learned weights encode source-conditioned relations, while current value features carry the evolving target appearance.

III. PRELIMINARIES

A. Latent Diffusion Editing

Let z_t be the latent at diffusion timestep t , and let $\epsilon_\theta(z_t, t, y)$ be the noise predicted under text condition y . Classifier-

free guidance [5] combines conditional and unconditional predictions as

$$\hat{\epsilon}_t = \epsilon_\theta(z_t, t, \emptyset) + s_{\text{cfg}} [\epsilon_\theta(z_t, t, y) - \epsilon_\theta(z_t, t, \emptyset)], \quad (1)$$

where s_{cfg} controls prompt strength. DDIM inversion maps the source image x^{src} to a timestep-aligned trajectory $\{z_{t,\text{src}}\}$, which allows source and edited representations to be compared at the same noise level.

B. Attention

For one self-attention head, hidden representations are projected into queries, keys, and values:

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V, \quad (2)$$

where d is the head dimension. Cross-attention has the same form but obtains keys and values from text embeddings. In our framework, cross-attention measures concept evidence, while self-attention supplies structural relations.

IV. METHOD

A. Overview

Fig. 2 separates the framework into structural-target generation, LSK fitting, and online editing. Given a source image x^{src} , we first compute a DDIM-inverted source trajectory. To generate structural targets for LSK training, we additionally construct a piecewise-constant semantic reference x^{sem} from the source image and invert it independently. Adaptive Style Nursing corrects concept deficits, LSSG aligns selected source features, and ESSG aligns informative self-attention relations with the semantic-reference trajectory. We collect the resulting source-anchored attention responses across images and timesteps and use them to train LSK.

At inference, LSK replaces the auxiliary semantic-reference branch at selected self-attention layers. The editor still uses the source trajectory for LSSG and sky/road masks for Adaptive Style Nursing, but it no longer inverts or evaluates x^{sem} . LSK predicts structural attention weights from the hidden states of the current forward branch, spatial coordinates, and the timestep, then aggregates the current value features. This separation preserves source-conditioned relations without freezing the target appearance.

B. Adaptive Style Nursing

Adaptive Style Nursing determines which concepts require reinforcement, where each concept should be expressed, and how strongly the latent should be updated.

1) *Concept Selection and Region Routing*: We parse the editing prompt into style-bearing token spans $\mathcal{T}_c$. Each concept is assigned a support according to its semantic role: global weather uses the full image, sky phenomena and illumination use the sky, and surface effects use the road. A frozen segmentation model [44] supplies sky and road masks, which are resized to the latent resolution to obtain M_c . This routing prevents every concept from being reinforced over the entire scene.

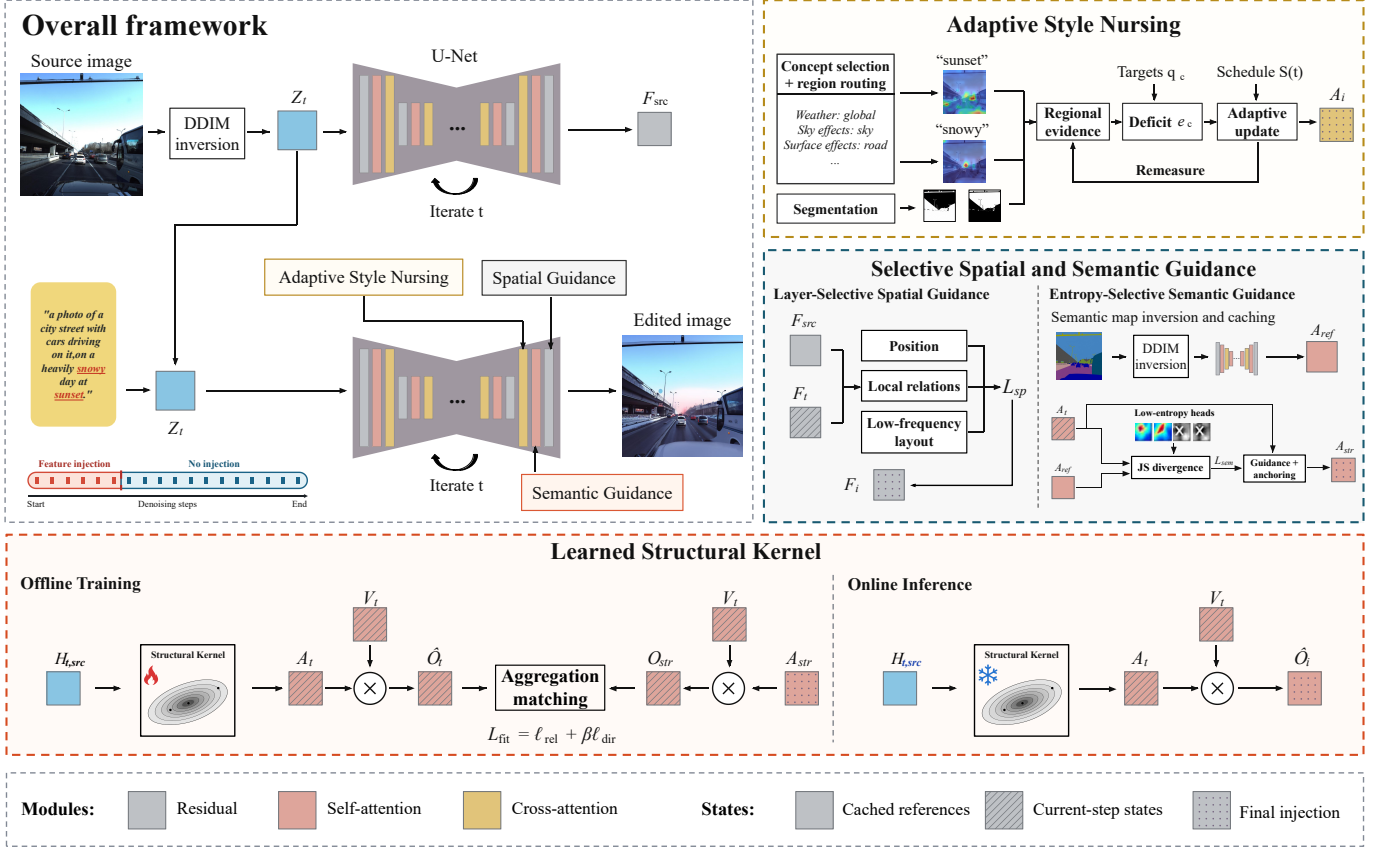


Fig. 2. Overview of *Style Nursing++*. DDIM inversion initializes editing and provides timestep-aligned source features. The editing U-Net follows the target prompt with style and structural guidance, with feature injection restricted to early denoising. Adaptive Style Nursing routes prompt concepts to the whole image, sky, or road, and updates the latent according to regional cross-attention deficits, remeasuring evidence after each correction. Layer-Selective Spatial Guidance compares current and source features to preserve position, local relations, and low-frequency layout. Entropy-Selective Semantic Guidance aligns current self-attention with A_{ref} on low-entropy reference heads using Jensen–Shannon divergence. The reference A_{ref} is cached along an independently inverted semantic-map trajectory; guidance and source anchoring produce structural attention A_{str} . During offline training, the Learned Structural Kernel (LSK) predicts current-hidden-state- and timestep-conditioned weights and learns to reproduce the aggregation $A_{str} V_t$ using the same current values V_t . During online inference, the frozen kernel aggregates current values at selected self-attention layers, replacing the auxiliary semantic-reference branch while retaining Adaptive Style Nursing and spatial guidance.

2) *Regional Concept Evidence*: For concept c , we aggregate cross-attention over selected layers, heads, and all tokens in its span:

$$a_c = \text{Smooth} \left(\text{Avg}_{\ell, h} \sum_{k \in \mathcal{T}_c} A_{t, \text{cross}}^{\ell h}(:, :, k) \right). \quad (3)$$

Within M_c , distributed concepts are evaluated by regional mean and coverage, whereas localized concepts use upper-tail and peak responses. Let $s_{c,j}$ be an observed statistic, $q_{c,j}$ its target, and $\omega_{c,j}$ its normalized weight. The remaining deficit is

$$e_c = \left[\sum_j \omega_{c,j} \left[1 - \frac{s_{c,j}}{q_{c,j}} \right]_+^2 \right]^{1/2}. \quad (4)$$

Only unmet targets contribute. Distributed weather is therefore reinforced until it covers enough of its support, while a localized concept such as lightning is encouraged to form a strong response within the sky.

3) *Deficit-Adaptive Update*: Let $\mathcal{L}_c = e_c^2$. The concept-specific size combines a base strength, an active denoising

schedule, the current deficit, and gradient normalization:

$$\eta_{c,t} = \bar{\eta}_c s_c(t) \frac{g(e_c)}{\|\nabla_{z_t} \mathcal{L}_c\|_2 + \epsilon}, \quad (5)$$

where $g(e_c)$ increases with the deficit and saturates for stability. The regional proposal is

$$\Delta z_{c,t} = -\eta_{c,t} M_c \odot \mathcal{S}_c(\nabla_{z_t} \mathcal{L}_c), \quad (6)$$

where $\mathcal{S}_c$ smooths corrections for distributed weather and illumination, while localized concepts retain concentrated gradients. Each update is limited by a per-step cap and a cumulative budget. We then recompute attention and deficit statistics, which stops further correction once the target has been met. Global concepts are updated before local concepts.

Fig. 3 illustrates the concept-specific schedules. The distributed rain concept is emphasized earlier, whereas the localized lightning target increases later in denoising.

C. Selective Spatial and Semantic Guidance

1) *Layer-Selective Spatial Guidance*: At timestep t , let F_t^ℓ be a feature map from the current edited latent, and let $F_{t,src}^\ell$ be the timestep-aligned reference from the source

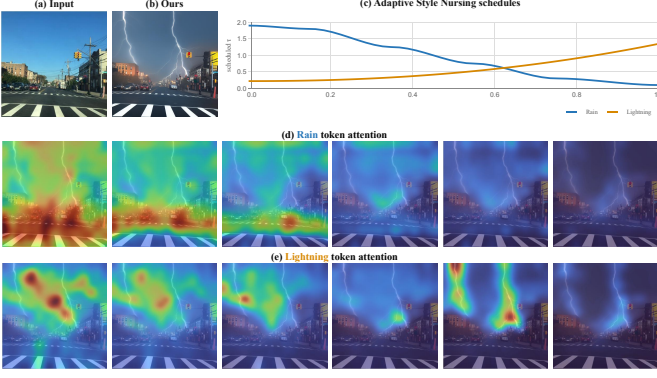


Fig. 3. Adaptive Style Nursing over denoising for a compound rain-and-lightning prompt. (a) Source image and (b) edited result. (c) Concept-specific target schedules emphasize distributed rain earlier and increase the localized lightning target later. (d)–(e) Cross-attention maps at sampled timesteps show the corresponding evolution of rain and lightning evidence. LSK is disabled in this diagnostic to isolate Adaptive Style Nursing.

trajectory. Both branches use the same content-only condition, so their discrepancy reflects structural change rather than direct activation of style tokens.

Different U-Net stages expose structure at different scales. We examine five representative stages at $t = 381$ for four traffic scenes and project the first three principal components of their residual features and self-attention relations to RGB (Fig. 4). Encoder 32×32 features contain fragmented local responses. Encoder 16×16 and Middle 8×8 form more continuous regions that expose coarse layout and foreground position. The bottleneck preserves large-scale organization but has limited boundary resolution. Decoder 16×16 restores local geometry while retaining regional continuity, and Decoder 32×32 adds finer contours and background detail.

We therefore apply spatial guidance at Encoder 16×16 , Middle 8×8 , and Decoder 16×16 . These stages jointly constrain coarse layout and local geometry while leaving high-resolution appearance synthesis less restricted.

We summarize the spatial discrepancy with three terms:

$$d_{\text{sp}}(F, G) = d_{\text{pos}}(F, G) + \lambda_{\text{nbr}} d_{\text{nbr}}(F, G) + \lambda_{\text{low}} d_{\text{low}}(F, G). \quad (7)$$

The positional term aligns feature directions at corresponding locations, the neighborhood term preserves local similarity patterns, and the low-frequency term anchors coarse layout. The layer-selective objective is

$$\mathcal{L}_{\text{sp}} = \sum_{\ell \in S_t} \bar{w}_{\ell} d_{\text{sp}}(F_t^{\ell}, F_{t, \text{src}}^{\ell}), \quad (8)$$

where S_t is the active layer set and $\bar{w}_{\ell}$ are normalized layer weights. The guidance is strongest early in denoising and is gradually relaxed as weather-specific details emerge.

2) *Entropy-Selective Semantic Guidance*: The semantic relation reference comes from a separate image stream. We construct x^{sem} from x^{src} by grouping appearance- and position-consistent pixels into coherent regions, then invert it to obtain a timestep-aligned trajectory. Let $A_{t, \text{sem}}^{\ell h}$ denote self-attention from this semantic reference and $A_t^{\ell h}$ self-attention from the current edited image. Both streams use the same content-only condition.

The self-attention projections in Fig. 4(b) exhibit smoother regional organization than the residual features. They capture structure- and semantics-related relations, including strong position dependence, but they are not category segmentations. ESSG therefore aligns selected relation distributions rather than treating PCA colors as labels.

Attention heads differ in selectivity. Diffuse heads encode nearly uniform relations, whereas concentrated heads identify more specific associations. For each reference head, we compute normalized entropy

$$H_{\ell h} = -\frac{1}{N \log N} \sum_{i=1}^N \sum_{j=1}^N A_{t, \text{sem}}^{\ell h}(i, j) \log A_{t, \text{sem}}^{\ell h}(i, j), \quad (9)$$

and retain the lower-entropy heads. The selected current and reference rows are aligned with Jensen–Shannon divergence:

$$\mathcal{L}_{\text{sem}} = \text{Avg}_{\ell} \sum_{h \in \mathcal{H}_{\ell, t}} \bar{v}_{\ell h} \text{JS}(A_t^{\ell h}, A_{t, \text{sem}}^{\ell h}). \quad (10)$$

Here, $\mathcal{H}_{\ell, t}$ is the selected head set and $\bar{v}_{\ell h}$ gives larger weight to more concentrated reference heads. The objective preserves object-internal and inter-object relations without forcing every head to copy the reference.

3) *Balanced Structural Update*: The spatial and semantic losses operate at different numerical scales. We normalize their latent gradients with running magnitude estimates before combining them:

$$\Delta z_t^{\text{str}} = -\gamma_{\text{sp}}(t) \frac{\nabla_{z_t} \mathcal{L}_{\text{sp}}}{m_{\text{sp}}(t) + \epsilon} - \gamma_{\text{sem}}(t) \frac{\nabla_{z_t} \mathcal{L}_{\text{sem}}}{m_{\text{sem}}(t) + \epsilon}. \quad (11)$$

The combined update is norm-bounded before application so that neither objective dominates because of raw gradient scale.

a) *Temporal scheduling*.: Fig. 5 visualizes Decoder 16×16 residual features and Decoder 32×32 self-attention relations at $t \in \{981, 581, 381, 1\}$. These layers and timesteps are diagnostic examples. Within each row, the PCA basis and color range are fixed across time. Early responses are smooth and coarse; as denoising proceeds, foreground regions and object boundaries become more distinct. From the middle to the late stages, their location and overall shape remain comparatively stable while local boundaries continue to sharpen.

This evolution motivates early structural anchoring followed by gradual release. The schedule $\gamma_{\text{sp}}(t)$ activates spatial guidance early, while $\gamma_{\text{sem}}(t)$ introduces relational constraints in stages. Both updates weaken and stop before the final denoising stage. In the full inference pipeline, LSK replaces the explicit semantic-reference update.

Fig. 6 provides an object-level diagnostic. PnP loses the source bus, while the combined structural guidance retains both the visible bus and source-aligned responses in residual features and bus-query self-attention.

D. Learned Structural Kernel

LSK learns the structural attention patterns produced by semantic guidance. Let $A_{t, \text{str}}^{h, b}$ denote the attention map from the semantic-guided, source-anchored trajectory at timestep t , head h , and forward branch b . We use these maps as structural targets and represent their weights with a normalized Gaussian kernel

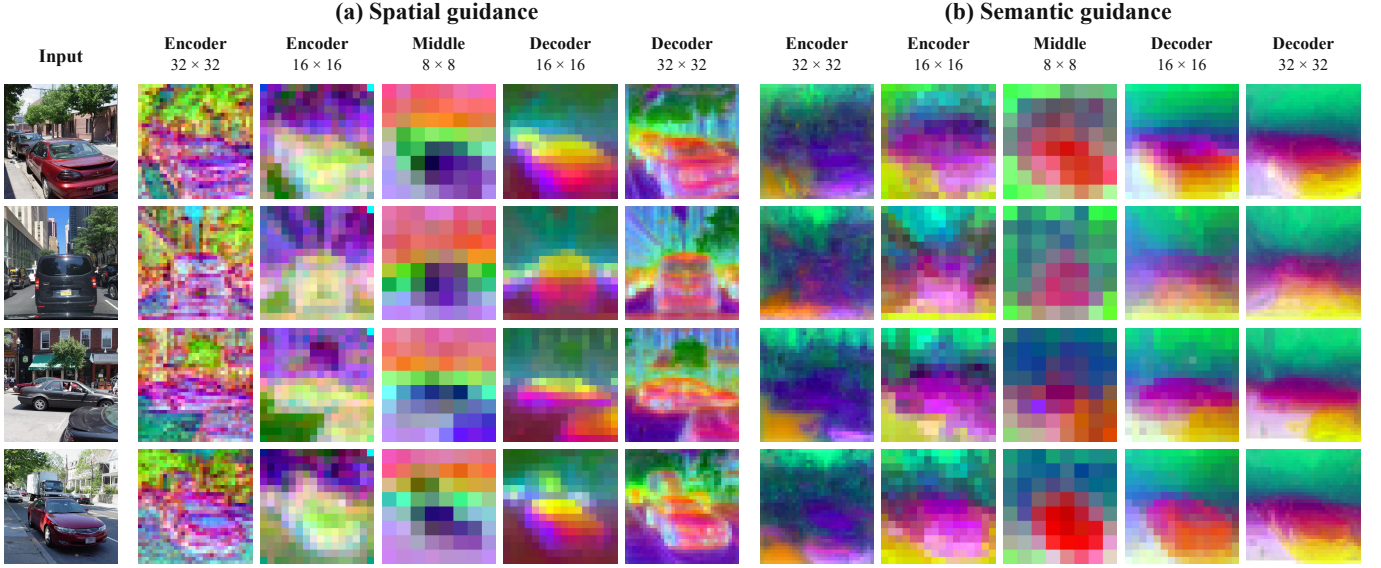


Fig. 4. Layer-wise representations used to select structural guidance stages across diffusion layers. Each row shows one traffic scene. RGB values are the first three principal components of (a) residual features and (b) self-attention relations at representative U-Net stages. Intermediate stages retain complementary coarse layout and local geometry, while the final high-resolution decoder recovers finer appearance detail. The PCA colors encode projection coordinates rather than semantic classes.

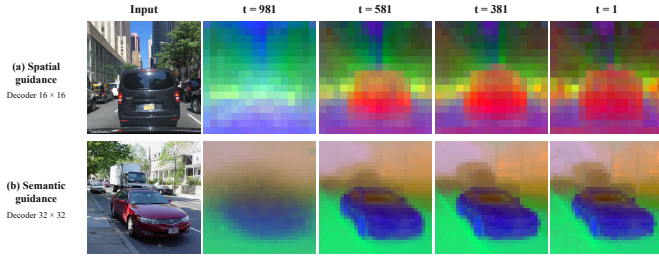


Fig. 5. Temporal evolution of structural representations during the denoising process. RGB PCA projections show (a) Decoder 16×16 residual features and (b) Decoder 32×32 self-attention relations for their respective source scenes. Each row uses a fixed projection basis and color range. Object-related regions become more distinct as noise decreases, while their location and overall shape stabilize from the middle to the late stages.

conditioned on current hidden representations and diffusion time.

1) *From Gaussian Kernel Regression to Attention*: Let k_j be a sample in feature space and q_i a query. Their Gaussian similarity is

$$\kappa_\sigma(q_i, k_j) = \exp\{-\|q_i - k_j\|_2^2 / (2\sigma^2)\}. \quad (12)$$

A weighted Nadaraya–Watson estimator normalizes these similarities as

$$\pi_{ij} = \frac{w_j \kappa_\sigma(q_i, k_j)}{\sum_m w_m \kappa_\sigma(q_i, k_m)}, \quad (13)$$

and aggregates associated values v_j :

$$\hat{o}_i = \sum_j \pi_{ij} v_j. \quad (14)$$

This normalized weighting and value aggregation has the same form as attention.

To make the equivalence explicit, set $\tilde{q}_i = q_i/d^{1/4}$ and $\tilde{k}_j = k_j/d^{1/4}$. Completing the square gives

$$\frac{q_i^\top k_j}{\sqrt{d}} = -\frac{\|\tilde{q}_i - \tilde{k}_j\|_2^2}{2} + \frac{\|\tilde{q}_i\|_2^2}{2} + \frac{\|\tilde{k}_j\|_2^2}{2}. \quad (15)$$

For a fixed query, the query-norm term cancels under row-wise softmax. The remaining key-dependent term defines

$$w_j = \exp\{\|\tilde{k}_j\|_2^2 / 2\}. \quad (16)$$

The attention coefficient can therefore be written as

$$\alpha_{ij} = \frac{w_j \kappa_1(\tilde{q}_i, \tilde{k}_j)}{\sum_m w_m \kappa_1(\tilde{q}_i, \tilde{k}_m)}. \quad (17)$$

Thus, scaled dot-product attention is a normalized weighted Gaussian kernel regressor. LSK learns this form from source-conditioned structural targets.

2) *Learning Structural Attention*: At each selected layer, we combine the current hidden token $h_{t,j}^b$, spatial coordinate r_j , and timestep embedding e_t . Here, b indexes the branch of the current U-Net forward pass, including a CFG branch or an auxiliary guidance branch; the layer index is omitted for clarity. The kernel does not read cached source hidden states:

$$u_{t,j}^b = [\text{LN}(h_{t,j}^b); r_j; e_t]. \quad (18)$$

For head h , the learned Gaussian log-affinity is

$$s_{t,ij}^{h,b} = -\frac{\|f_q^h(u_{t,i}^b) - f_k^h(u_{t,j}^b)\|_2^2}{2\sigma_{t,h}^2} + \omega_{t,j}^{h,b}, \quad (19)$$

where f_q^h and f_k^h are learned mappings, $\sigma_{t,h} > 0$ is the bandwidth, and $\omega_{t,j}^{h,b}$ is a learned log-weight computed from the current token representation. Row normalization gives the Structural Kernel

$$K_{t,ij}^{h,b} = \frac{\exp\{s_{t,ij}^{h,b}\}}{\sum_m \exp\{s_{t,im}^{h,b}\}}. \quad (20)$$

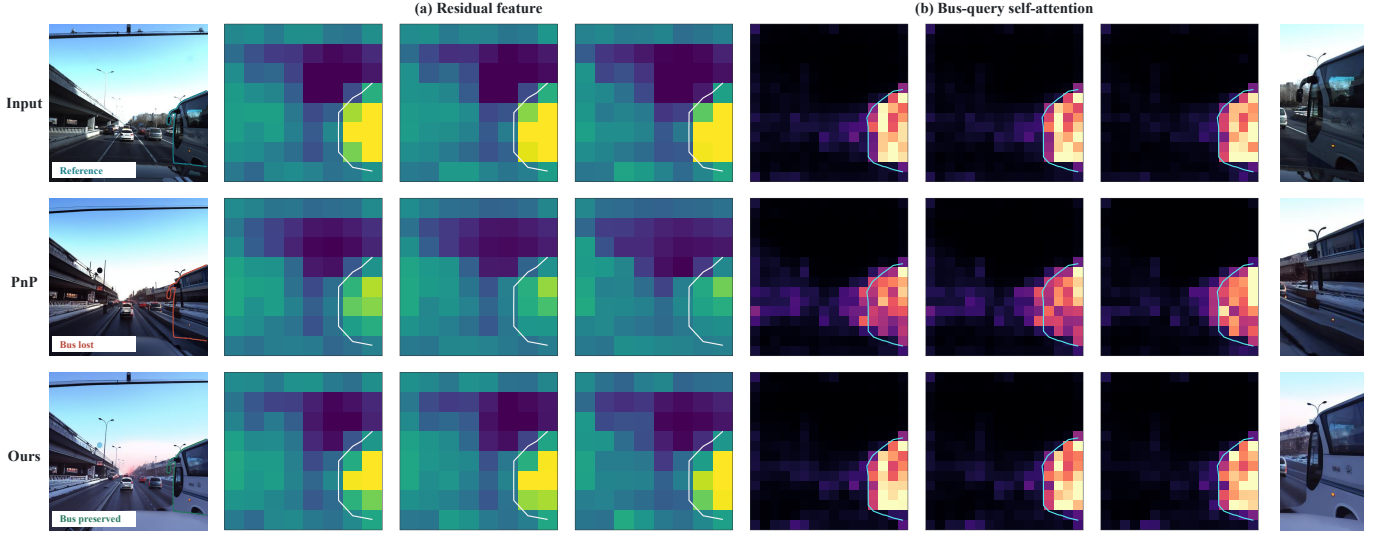


Fig. 6. Structural diagnostics for the bus region. Rows show the source, Plug-and-Play (PnP), and *Style Nursing++*. The outer image columns show that PnP removes the bus, whereas *Style Nursing++* preserves it. (a) Residual-feature projections and (b) bus-query self-attention maps show the corresponding internal responses. White contours indicate the bus support in the source.

Each row of $K_t^{h,b}$ is a distribution over spatial positions. The mappings, bandwidth, and log-weights adapt that distribution to the current hidden representations and denoising time. Source information can affect these representations through the surrounding editing pipeline, but cached source hidden states are not direct inputs to LSK.

We supervise the kernel through its aggregation of current value features. Applying the target map and learned kernel to the same $V_t^{h,b}$ gives

$$O_{t,\text{str}}^{h,b} = A_{t,\text{str}}^{h,b} V_t^{h,b}, \quad (21)$$

$$\hat{O}_t^{h,b} = K_t^{h,b} V_t^{h,b}. \quad (22)$$

Using the same values isolates the structural aggregation that LSK must reproduce. Training examples span source images, timesteps, selected layers, heads, and CFG branches. Denoting this collection by $\mathcal{D}_{\text{str}}$, we optimize

$$\mathcal{L}_{\text{fit}} = \mathbb{E}_{\mathcal{D}_{\text{str}}} \left[\frac{\|\hat{O} - O_{\text{str}}\|_F^2}{\|O_{\text{str}}\|_F^2 + \epsilon} + \beta(1 - \cos(\hat{O}, O_{\text{str}})) \right]. \quad (23)$$

The relative reconstruction and cosine terms match both magnitude and direction.

3) *Inference with the Structural Kernel*: At inference, all LSK parameters are frozen. At each selected layer ℓ , the kernel computes weights from the hidden states of the current branch and the timestep, and aggregates the current values:

$$\hat{O}_t^{\ell,b} = K_\phi^\ell(H_t^{\ell,b}, t) V_t^{\ell,b}. \quad (24)$$

The current hidden states determine the structural weights, while the current values carry the evolving appearance of the corresponding branch. Computing these weights does not require reading cached source or reference Q/K tensors. This input definition alone does not imply removal of other reference-dependent guidance in the surrounding pipeline. LSK replaces explicit semantic-reference guidance at the selected layers; Adaptive Style Nursing continues to use sky and road masks for regional control.

E. Overall Inference

At each denoising step, Adaptive Style Nursing measures global and local concept deficits and applies bounded regional corrections. LSSG then anchors the latent to timestep-matched source features. At selected self-attention layers, LSK predicts structural weights from the current branch hidden states and timestep and applies them to the current values. CFG and the DDIM scheduler then advance the latent. The complete trajectory combines adaptive concept realization, source-feature anchoring, and learned structural aggregation.

V. EXPERIMENTS

A. Experimental Setup

a) *Main evaluation.*: We evaluate compound and single-weather editing on STREET-6K. The comparison includes SDEdit, Prompt-to-Prompt with Null-text inversion (P2P+NTI), PnP, Pix2Pix-Zero, MasaCtrl, DiffuseIT, LEDITS++, Free-Prompt-Editing, and Style Nursing. We report eleven complementary measures. DINO and SAM dense cosine similarities quantify structural agreement [45], [46]; PSNR, LPIPS [47], MSE, and SSIM [48] measure source reconstruction; CLIP image, text-image, and directional similarities [49] measure visual and semantic alignment; and EditScore provides its SC and Overall outputs [50]. PSNR is reported in decibels, MSE in units of 10^{-4} , and the remaining values are dimensionless.

We distinguish two configurations. The *guidance-only core* uses Adaptive Style Nursing, LSSG, and explicit ESSG. The *full model* retains Adaptive Style Nursing and LSSG but replaces the auxiliary ESSG trajectory with LSK at inference. Component ablations are applied to the guidance-only core unless stated otherwise.

b) *LSK fitting evaluation.*: The fitting study uses all 4,320 held-out traces and averages the conditional and unconditional CFG branches. Relative MSE is the normalized squared reconstruction error used in the fitting objective. MAE and NRMSE measure absolute and normalized errors, R^2 and

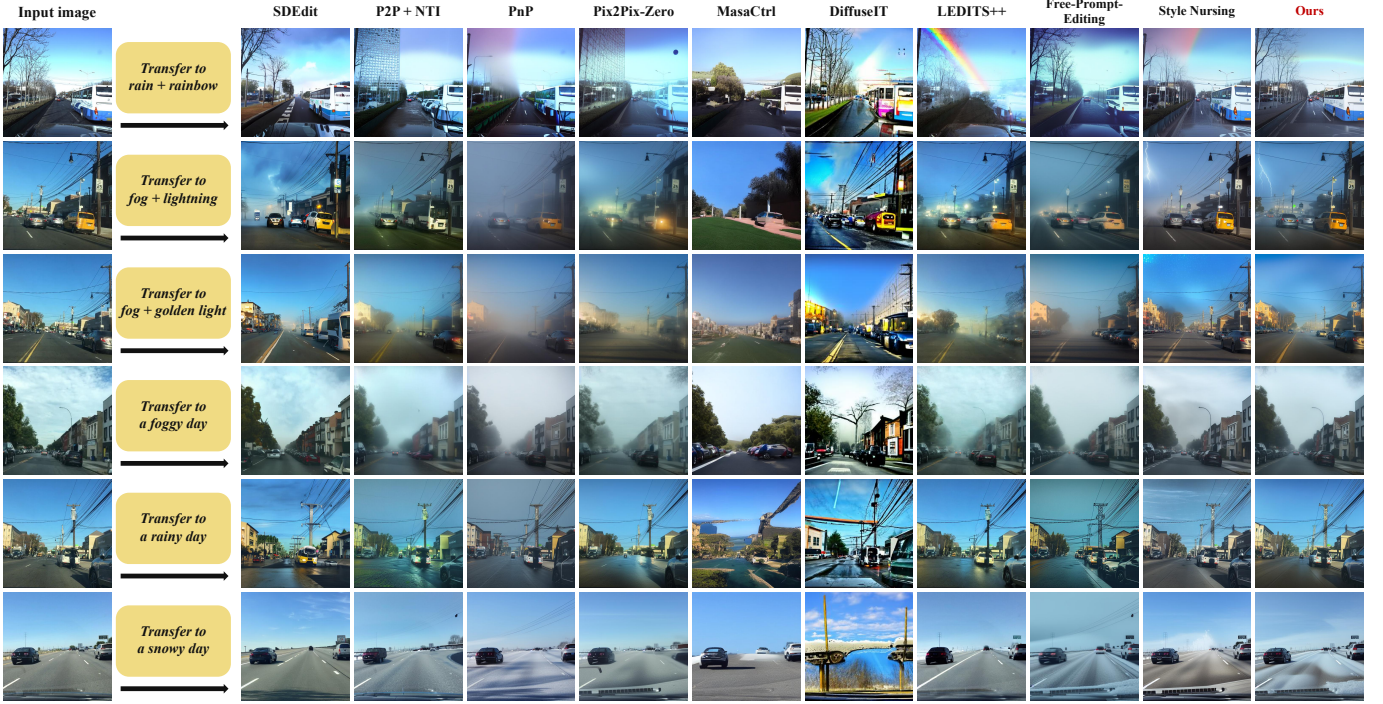


Fig. 7. Qualitative comparison on compound and single-weather transfer. Columns show the source image, requested prompt, SDEdit, P2P+NTI, PnP, Pix2Pix-Zero, MasaCtrl, DiffuseIT, LEDITS++, Free-Prompt-Editing, Style Nursing, and the full *Style Nursing++* model. Rows request rain + rainbow, fog + lightning, fog + golden light, fog, rain, and snow. The comparison exposes both concept realization and preservation of source vehicles and scene layout.

Pearson correlation measure agreement, and Cosine Error measures directional mismatch.

c) *Efficiency protocol.*: Runtime is measured on an NVIDIA RTX PRO 6000 Blackwell Server Edition using 100 STREET-6K images at 512×512 resolution. Both variants use the same prompt, CFG setting, corresponding seed, and 50 diffusion steps. Timing starts after model loading and includes source inversion, semantic-reference inversion for the explicit-guidance core, guidance, denoising, and image saving.

B. Main Comparison

1) *Qualitative Results*: Fig. 7 compares six traffic scenes: three compound prompts (rain + rainbow, fog + lightning, and fog + golden light) and three single-weather prompts (fog, rain, and snow). Methods that remain close to the input often under-express one of the requested effects, whereas methods with stronger appearance changes may alter vehicles, buildings, or road geometry. Across these examples, the full model realizes the target weather while retaining the main source layout and objects.

2) *Quantitative Results*: Table I shows that no baseline dominates every objective. Style Nursing gives the highest DINO, SAM, and SSIM; LEDITS++ gives the lowest LPIPS and highest CLIP-I; DiffuseIT gives the highest CLIP-TI; and PnP gives the highest EditScore Overall. The full *Style Nursing++* model achieves the best PSNR, MSE, CLIP-Dir, and EditScore SC. It is second in DINO, SAM, LPIPS, and EditScore Overall. This profile places the full model near the leading preservation methods while maintaining strong directional and edit-specific alignment.

Relative to the guidance-only core, LSK improves every reported metric: DINO/SAM rise from 0.7580/0.8023 to 0.7600/0.8057; PSNR rises from 20.04 dB to 20.16 dB; LPIPS decreases from 0.2407 to 0.2381; and MSE decreases from 111.52 to 108.81. The five semantic and reward measures also increase.

C. Component Analysis

Table II clarifies the role of each component. Removing Adaptive Style Nursing shifts the model toward source reconstruction: preservation metrics improve, but CLIP-TI, CLIP-Dir, and both EditScore outputs decrease substantially. Removing LSSG weakens all structural and reconstruction measures and also lowers CLIP-I. Removing ESSG produces smaller, mixed changes in reconstruction, but it reduces SAM, worsens LPIPS, and lowers CLIP-I, CLIP-Dir, and both EditScore outputs. Replacing explicit ESSG with LSK improves all eleven metrics over the core.

Fig. 8 provides the visual counterpart. Without Adaptive Style Nursing, the requested compound effects are weak. Removing LSSG or ESSG changes the balance between weather realization and preservation of vehicles and scene layout.

D. Learned Structural Kernel Evaluation

1) *Fitting Architecture*: Under the same split and training settings with approximately matched parameter counts, the Gaussian kernel is best on all six metrics in Table III. Its Relative MSE is 20.5 % lower than the MLP dot-product model and 51.3 % lower than the convolutional dot-product model.

TABLE I

MAIN QUANTITATIVE COMPARISON ON STREET-6K. DINO AND SAM ARE DENSE COSINE STRUCTURE SIMILARITIES. CLIP-I, CLIP-TI, AND CLIP-Dir DENOTE IMAGE, TEXT-IMAGE, AND DIRECTIONAL SIMILARITIES; SC AND OVERALL ARE THE TWO EDITSCORE OUTPUTS. PSNR IS REPORTED IN DECIBELS, MSE IN UNITS OF 10^{-4} , AND THE REMAINING COLUMNS ARE DIMENSIONLESS. THE GUIDANCE-ONLY CORE USES EXPLICIT ESSG, WHEREAS THE FULL MODEL USES LSK AT INFERENCE. BEST AND SECOND-BEST VALUES IN THIS TABLE ARE BOLD AND UNDERLINED.

Method	DINO ↑	SAM ↑	PSNR (dB) ↑	LPIPS ↓	MSE (10^{-4}) ↓	SSIM ↑	CLIP-I ↑	CLIP-TI ↑	CLIP-Dir ↑	EditScore SC ↑	EditScore Overall ↑
SDEdit	0.6436	0.6752	19.79	0.2958	121.35	0.7001	0.7650	0.2326	0.1285	5.882	6.189
P2P + NTI	0.7573	0.7964	19.74	0.2515	120.42	0.7852	<u>0.7819</u>	0.2578	0.1906	6.706	6.796
PnP	0.7150	0.7681	18.18	0.3081	173.52	0.7572	0.7340	<u>0.2687</u>	0.1971	7.392	7.730
Pix2Pix-Zero	0.7434	0.7878	19.78	0.2642	116.95	0.7760	0.7801	0.2446	0.1476	5.912	5.754
MasaCtrl	0.4506	0.5882	16.01	0.4093	272.08	0.6731	0.6346	0.1795	0.0205	1.376	1.561
DiffuseIT	0.5663	0.5929	13.80	0.4854	442.78	0.4800	0.7154	0.2822	0.1714	5.064	4.308
LEDITS++	0.7420	0.7611	19.64	0.2377	120.49	<u>0.7854</u>	0.7878	0.2585	<u>0.2002</u>	6.956	6.972
Free-Prompt-Editing	0.6871	0.7257	11.36	0.3902	755.01	0.6673	0.7235	0.2666	0.1870	6.302	7.166
Style Nursing (SN)	0.7770	0.8091	19.49	0.2389	121.42	0.7857	0.7814	0.2557	0.1967	7.510	7.144
<i>Style Nursing++ core (explicit ESSG)</i>	0.7580	0.8023	<u>20.04</u>	0.2407	<u>111.52</u>	0.7814	0.7626	0.2605	0.1965	<u>7.878</u>	7.368
<i>Style Nursing++ full (LSK)</i>	<u>0.7600</u>	<u>0.8057</u>	20.16	<u>0.2381</u>	108.81	0.7839	0.7631	0.2618	0.2005	7.904	<u>7.396</u>

TABLE II

COMPONENT ANALYSIS ON STREET-6K. THE CORE USES ADAPTIVE STYLE NURSING, LSSG, AND EXPLICIT ESSG. EACH “w/o” ROW REMOVES ONE COMPONENT FROM THAT CORE. THE FINAL ROW REPLACES EXPLICIT ESSG WITH LSK AT INFERENCE. WE OMIT BEST/SECOND-BEST HIGHLIGHTING BECAUSE THESE ROWS REPRESENT DIFFERENT PRESERVATION-EDITING OPERATING POINTS.

Configuration	DINO ↑	SAM ↑	PSNR (dB) ↑	LPIPS ↓	MSE (10^{-4}) ↓	SSIM ↑	CLIP-I ↑	CLIP-TI ↑	CLIP-Dir ↑	EditScore SC ↑	EditScore Overall ↑
Core (ASN + LSSG + ESSG)	0.7580	0.8023	20.04	0.2407	111.52	0.7814	0.7626	0.2605	0.1965	7.878	7.368
w/o Adaptive Style Nursing	0.7923	0.8134	22.58	0.1841	61.35	0.8104	0.8234	0.2142	0.1154	5.592	6.071
w/o LSSG	0.7486	0.7958	19.04	0.2652	137.55	0.7696	0.7483	0.2627	0.1926	7.746	7.350
w/o ESSG	0.7549	0.7952	20.08	0.2439	110.76	0.7815	0.7549	0.2615	0.1923	7.680	7.067
Full (ASN + LSSG + LSK)	0.7600	0.8057	20.16	0.2381	108.81	0.7839	0.7631	0.2618	0.2005	7.904	7.396

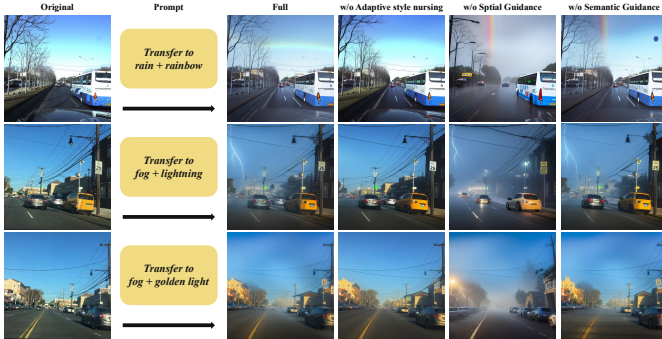


Fig. 8. Qualitative component analysis on three compound prompts. Rows request rain + rainbow, fog + lightning, and fog + golden light. Columns show the source, prompt, full model, and variants without Adaptive Style Nursing, LSSG, or ESSG. Removing Adaptive Style Nursing weakens the requested effects; removing either structural component changes object and layout preservation.

TABLE III

HELD-OUT COMPARISON OF ALTERNATIVE LSK FITTING ARCHITECTURES. RESULTS ARE AVERAGED OVER CONDITIONAL AND UNCONDITIONAL BRANCHES.

Method	Relative MSE ↓	MAE (response units) ↓	NRMSE ↓	R^2 ↑	Pearson ↑	Cosine Error ↓
Gaussian kernel	0.0210	0.0294	0.1245	0.9790	0.9896	0.0113
MLP dot-product	0.0264	0.0332	0.1407	0.9736	0.9868	0.0143
Conv dot-product	0.0431	0.0437	0.1847	0.9569	0.9781	0.0240

TABLE IV

HELD-OUT COMPARISON OF ALTERNATIVE KERNEL FUNCTIONS UNDER THE SAME LSK ARCHITECTURE AND TRAINING SETTINGS.

Kernel	Relative MSE ↓	MAE (response units) ↓	NRMSE ↓	R^2 ↑	Pearson ↑	Cosine Error ↓
Gaussian	0.0210	0.0294	0.1245	0.9790	0.9896	0.0113
Laplacian	<u>0.0213</u>	<u>0.0300</u>	<u>0.1264</u>	<u>0.9787</u>	<u>0.9894</u>	<u>0.0117</u>
Epanechnikov	0.0265	0.0335	0.1415	0.9735	0.9866	0.0146
Cauchy	0.0497	0.0481	0.2001	0.9503	0.9752	0.0279
Inverse multiquadric	0.0997	0.0687	0.2883	0.9002	0.9494	0.0553

2) *Kernel Function*: Gaussian and Laplacian kernels are close in Table IV: their Relative MSE differs by 1.4 %. Gaussian gives the lowest test error, whereas the predetermined validation criterion preferred Laplacian. We therefore treat their ranking as a near tie and avoid attributing the overall LSK gains to this small difference alone.

3) *Agreement with Transformer Self-Attention*: Fig. 9 compares LSK and native Transformer attention in space and after aggregation. For fixed input features and timestep, the learned kernel follows the principal response regions, and the column-wise attention-mass curves have similar peak locations and amplitudes. Across three evaluated examples and all query positions, mean-attention cosine similarity ranges from 0.9295 to 0.9787. With the same value matrix V , the cosine similarity between AV and $\hat{A}V$ ranges from 0.9737 to 0.9932.

Fig. 10 gives a single-example analysis at layer up2_attn0 and diffusion-step index 12. With fixed inputs and the same V , mean-head attention cosine similarity is 0.9619. The aggregated

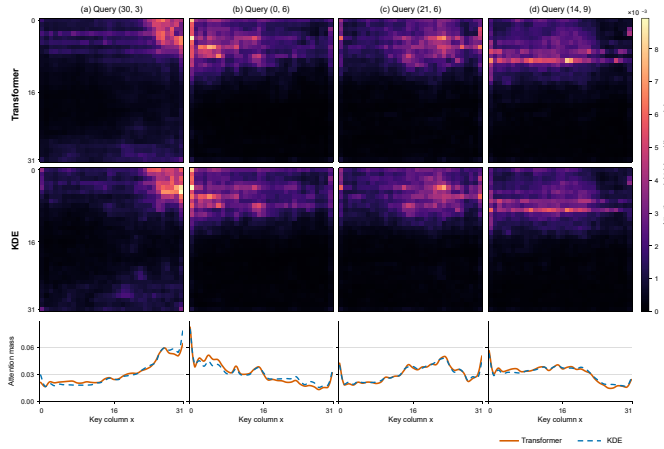


Fig. 9. Spatial agreement between native Transformer self-attention and LSK for four query positions. The first two rows show attention maps under a shared linear scale; the bottom row compares column-wise attention mass. The learned kernel recovers the main response regions and peak structure.

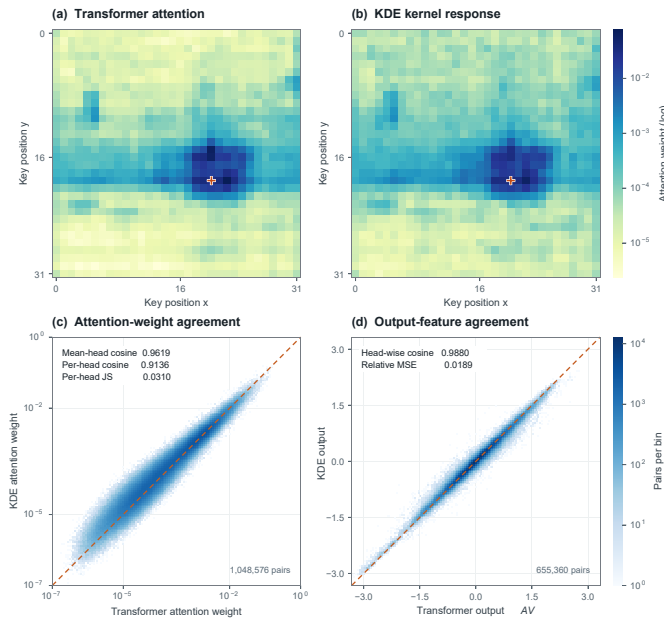


Fig. 10. Detailed LSK approximation at a representative decoder attention layer. (a)–(b) Native self-attention and LSK weights are visualized under a shared scale. (c)–(d) The comparison of attention weights and corresponding outputs demonstrates that the learned kernel captures the structural response of self-attention. Dashed diagonals indicate exact agreement.

TABLE V

PER-IMAGE EFFICIENCY COMPARISON ON STREET-6K USING AN NVIDIA RTX PRO 6000 BLACKWELL SERVER EDITION. LATENCY, COMPUTE, AND ENERGY CONSUMPTION ARE REPORTED IN THE CORRESPONDING EFFICIENCY METRICS.

Method	Mean latency (s/image)	Compute (TFLOP/image)	Energy (W h/image)	P50 latency (s/image)	P95 latency (s/image)
<i>Style Nursing++</i> core (explicit ESSG)	27.61	183.75	1.92	27.55	28.97
<i>Style Nursing++</i> full (LSK)	19.65	145.37	1.49	19.53	21.14

outputs reach head-wise cosine similarity 0.9880 with Relative MSE 0.0189.

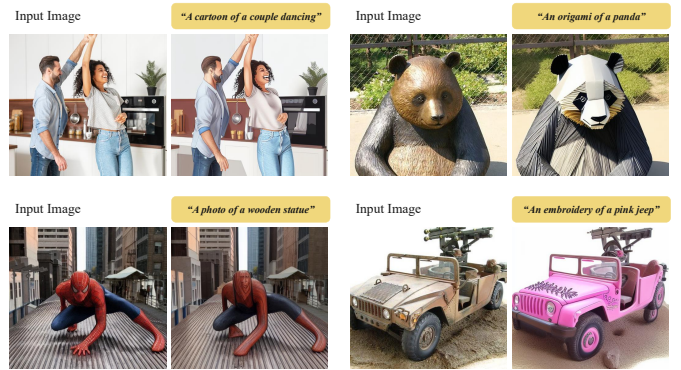


Fig. 11. Cross-domain edits produced with an LSK trained on traffic scenes. Each pair shows a source and edited result: a cartoon rendering of a dancing couple, an origami panda, a wooden statue, and an embroidered pink jeep. The edits alter appearance while retaining the dominant pose and spatial organization.

E. Computational Efficiency

Table V shows that LSK reduces mean latency by 28.8 %, compute by 20.9 %, and energy by 22.4 % relative to explicit ESSG. Median latency decreases from 27.55 s/image to 19.53 s/image, and P95 latency decreases from 28.97 s/image to 21.14 s/image.

F. Cross-Domain Transfer Examples

Fig. 11 applies the traffic-trained LSK to object editing outside traffic scenes. The examples change rendering style, material, or object appearance while retaining the dominant pose, silhouette, and spatial arrangement. They provide qualitative evidence of cross-domain transfer; a broader benchmark is needed for quantitative generalization claims.

VI. DISCUSSION

The results reveal a preservation–editing trade-off rather than a single scalar ranking. Removing Adaptive Style Nursing moves the system toward reconstruction and substantially weakens prompt realization. Removing LSSG degrades geometry and reconstruction, while removing ESSG weakens several relational, perceptual, and edit-specific measures. Among complete methods, the full model combines leading PSNR, MSE, CLIP-Dir, and EditScore SC with competitive DINO, SAM, and LPIPS. LSK further improves every metric over the explicit-guidance core while lowering inference cost. These complementary effects support the intended division of labor: Adaptive Style Nursing supplies concept activation, LSSG anchors multiscale geometry, ESSG provides relational targets, and LSK transfers those relations to efficient inference.

The fitting experiments support the Gaussian structural parameterization over matched dot-product alternatives. The comparison between Gaussian and Laplacian kernels is less decisive because their errors are close and their validation and test rankings differ. The evidence therefore supports the learned kernel formulation more strongly than a unique choice of kernel family.

The current region router covers supports that can be obtained reliably from traffic scenes: the full image, sky, and road.

New localized concepts require additional masks or a learned routing policy. LSK is fitted to selected layers and denoising stages of a particular diffusion backbone, so a substantially different backbone requires new structural traces and refitting. The efficiency study uses one GPU and one resolution, and the cross-domain study is qualitative. Evaluating additional hardware, backbones, and domains would characterize the portability of the measured gains.

VII. CONCLUSION

We presented *Style Nursing++* for zero-shot compound weather transfer. Adaptive Style Nursing routes prompt concepts to compatible regions and reinforces them according to regional deficits. LSSG preserves multiscale geometry, and ESSG supplies informative relation targets through entropy-selected self-attention. LSK learns these current-hidden-state- and timestep-conditioned structural responses and replaces the auxiliary semantic-reference trajectory at inference. On STREET-6K, LSK improves all eleven reported metrics over the guidance-only core while reducing mean latency by 28.8%, compute by 20.9%, and energy by 22.4%. The results demonstrate that compound weather editing benefits from coordinating concept realization and structural preservation throughout denoising.

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